

Education

- 2020– Now **Ph.D. in Electrical Engineering (Communications)**, *University of Toronto*, Toronto
Total GPA: 3.86/4 via 2.5 credits
- 2017–2020 **M.Sc. in Electrical Engineering (Communication Networks)**, *University of Tehran*, Tehran
Total GPA: 17.39/20 via 32 credits
- 2013–2017 **B.Sc. in Electrical Engineering (Communications)**, *University of Tehran*, Tehran
Total GPA: 3.53/4 via 107 credits
Junior Year GPA: 4/4 or 19.17/20 via 34 credits

Honors and Awards

- 2024 Winner (ranked 1st) of ITU AI/ML in 5G Challenge.
- 2020-2025 Student Fellowship, University of Toronto.
- 2020-2025 Research Fellowship, University of Toronto.
- Fall 2020 Graduated top of the class in M.Sc. Communication students of University of Tehran.
- Fall 2017 University of Tehran exceptional talented student in B.Sc.
- Summer 2017 Graduated top of the class in B.Sc. Electrical Engineering students of University of Tehran.
- 2016 An honorary member of **University of Tehran Elites Foundation**.

Power System Related Projects

Static Compensator (STATCOM)

Implemented a three-level VSC-based STATCOM in PSCAD; performed dynamic and steady-state studies of reactive-power support and voltage regulation under grid disturbances.

Back-to-Back VSC-HVDC Link

Built a ± 320 kV back-to-back HVDC model in PSCAD; analyzed active-power transfer, DC-voltage control, and fault ride-through performance.

Variable-Speed Wind-Power System

Modeled a DFIG-based wind turbine with VSC converters in PSCAD; evaluated maximum-power-point tracking, grid-code compliance, and low-voltage ride-through behavior.

Comprehensive Multi-Bus Power-System Study

Built a detailed PSCAD model of an interconnected transmission network (generators, transformers, lines, and dynamic loads); carried out load-flow calculations, extensive fault analysis (three-phase and single-line-to-ground), protection coordination with relay/breaker settings, and transient-stability simulations to assess system response to severe disturbances and validate analytical results.

Simple SCADA System Using Ignition

Implemented a Modbus TCP-based VSC simulation in Python (PyModbus, threading) integrated with Ignition SCADA and a custom HMI in Designer.

Skills

2013-2025 Power Systems

Standardized Communication Protocol for Substation Automation (IEC 61850), SV, GOOSE, and Modbus Protocols, Power Electronics (Inverters, Rectifiers, AC/AC and DC/DC Converters, and etc.), Power System Fundamentals (Transmission Lines, Faults, System Protection, Power Distribution System), IEEE Standard for Interconnection and Interoperability of Inverter-Based Resources (IBRs) Interconnecting with Associated Transmission Electric Power Systems (IEEE Std 2800).

- 2013-2025 **Communications and Networking (Major Field in Academia)**
Comprehensive knowledge in communication systems, wireless communications, wireless and computer networks (21 times TAs of Computer Networks Course), and cellular networks. OSI protocol stack (TCP/UDP, IP, link layer protocols, MAC layer protocols, etc.), traffic engineering, routing algorithms, SDN, distributed systems, Wireless Resource allocation, RRM, scheduling, etc..
- 2013-2025 **Machine Learning**
Deep Reinforcement learning techniques (DDPG, SAC, DQN, etc.), deep neural networks architectures (CNN, RNN, LSTM, Attention module, Transformers and etc.), Autoencoders (AE, VAE, SAE), Generative Adversarial Network (GAN), Graph Neural Network (GNN), Diffusion Models.
- 2013-2025 **Optimization**
Convex and non-convex optimization, duality theory, KKT conditions, linear and quadratic programming, mixed-integer programming, multi-objective optimization, stochastic and robust optimization, gradient-based and gradient-free methods, heuristic algorithms, ADMM, dynamic programming, interior-point methods, branch-and-bound, Bayesian optimization.
- 2013-2025 **Electrical Engineering**
Strong academic background and coursework in Electrical Engineering and Power Systems.

Computer Skills

- 2013-2024 **Programming Languages**
Python, C, C++, Socket Programming, MATLAB and Simulink, Pytorch, Tensorflow, MPI, R, AMPL, CVXPY, Gurobi, CPLEX, MATLAB Optimization Toolbox, SciPy optimization, parallel and distributed optimization.
- 2013-2024 **Industrial Tools**
PSCAD, Ignition SCADA, Hspice, Pspice, Multisim

Publications

H. Bijanrostami, E. Sousa and M. K. Farshbafan, "Goal-Oriented Semantic Communication for DNN Task Inference in IoT Networks," *2024 IEEE International Conference on Communications Workshops (ICC Workshops)*, Denver, CO, USA, 2024, pp. 1660-1665, doi: 10.1109/ICCWorkshops59551.2024.10615907.

Published

H. Bijanrostami, E. Sousa and M. K. Farshbafan, "Hierarchical Feature Encoding in 6G in-X Subnetworks Using Information Bottleneck," *2025 IEEE International Symposium on Personal, Indoor and Mobile Radio Communications*, Istanbul, Turkey.

Accepted

H. Bijanrostami, E. Sousa and M. K. Farshbafan, "Modular Information Bottleneck Encoding for Scalable Task Inference in IoT Networks," *2025 IEEE Global Communications Conference*, Taipei, Taiwan.

Accepted

H. Bijanrostami, E. Sousa, "Reinforcement Learning Based Task-Aware Resource Management for 6G In-X Subnetworks," *2026 IEEE International Conference on Communications*, Glasgow, Scotland.

Submitted

H. Bijanrostami, E. Sousa, "Multi-Task Two-Step Feature Encoding Using Information Bottleneck in Next-Generation Networks," *2026 IEEE International Conference on Communications*, Glasgow, Scotland.

Submitted

S. Al-kokhon, H. Bijanrostami and E. Sousa, "Optimal Multi-user scheduling in massive MIMO mobile channels using Soft Actor-Critic Reinforcement Learning", *ITU Journal on Future and Evolving Technologies (ITU J-FET)*.

Published (ITU Competition Winner (Ranked 1st))

H. Bijanrostami and E. Sousa, *"Machine Learning Based Semantic Communication Using Information Bottleneck Principle in Future Wireless Networks"*, Thesis Chapter and Journal Paper.

To be Submitted

H. Bijanrostami and E. Sousa, *"Energy-Efficient Memory-Constrained Edge Inference in Scalable IoT Networks Using Modular Information Bottleneck Encoding"*, Thesis Chapter and Journal Paper.

To be Submitted

H. Bijanrostami and E. Sousa, *"Universal Semantic Communication Framework for Task-Aware Inference in Multi-Tier In-X 6G Networks"*, Thesis Chapter and Journal Paper.

To be Submitted

H. Bijanrostami, *"Resource Allocation in Hierarchical Edge Computing IoT Networks Using Optimal Routing."*

Master's Thesis